

Part 1: Backend Fundamentals (Quick Revision)

1. What is Backend?

Interview Answer: Backend is the server-side part of an application. It handles business logic, authentication, database operations, and sends responses to the frontend.

2. What is Frontend?

Interview Answer: Frontend is the user interface of an application. It is built using HTML, CSS, and JavaScript and allows users to interact with the application.

3. Frontend vs Backend

Frontend	Backend
User Interface	Business Logic
Runs in Browser	Runs on Server
HTML, CSS, JS	Node.js, Java, Python

4. What is Full Stack?

Interview Answer: A Full Stack Developer works on both the frontend and backend of an application.

5. What is Client?

Interview Answer: A client is a browser, mobile app, or device that sends requests to a server and displays the server's response.

6. What is Server?

Interview Answer: A server is a computer or software that receives client requests, processes them, and sends responses back.

7. Client vs Server

Client	Server
Sends Request	Processes Request
Browser/App	Backend
Displays Data	Returns Data

8. How do Client and Server communicate?

Interview Answer: A client sends an HTTP request to the server. The server processes it, accesses the database if needed, and returns an HTTP response.

9. What is an API?

Interview Answer: An API (Application Programming Interface) allows different software applications to communicate and exchange data.

10. What is REST API?

Interview Answer: A REST API follows REST principles and uses HTTP methods like GET, POST, PUT, PATCH, and DELETE to perform CRUD operations.

11. What is HTTP?

Interview Answer: HTTP (HyperText Transfer Protocol) is the protocol used for communication between clients and servers.

12. What is HTTPS?

Interview Answer: HTTPS is the secure version of HTTP. It encrypts data using SSL/TLS to protect information during transmission.

13. What is a Request?

Interview Answer: A request is a message sent by the client to the server asking for data or an action.

14. What is a Response?

Interview Answer: A response is the data or result sent by the server back to the client after processing a request.

15. What is CRUD?

Interview Answer: CRUD stands for Create, Read, Update, and Delete, which are the four basic database operations.

16. What is JSON?

Interview Answer: JSON (JavaScript Object Notation) is a lightweight format used to exchange data between the frontend and backend.

17. What is XML?

Interview Answer: XML (eXtensible Markup Language) is a markup language used to store and transfer data, but today JSON is more commonly used in web applications.

18. What is a Database?

Interview Answer: A database is a system used to store, organize, manage, and retrieve data efficiently.

19. Types of Databases

Interview Answer: There are two main types of databases:

- SQL databases (MySQL, PostgreSQL)
- NoSQL databases (MongoDB, Cassandra)

20. SQL vs NoSQL

SQL	NoSQL
Tables	Documents
Fixed Schema	Flexible Schema
MySQL	MongoDB

21. What happens when you type `google.com`?

Interview Answer: The browser finds Google's IP address using DNS, sends an HTTPS request to the server, the server processes it and returns HTML, CSS, and JavaScript, which the browser renders as the webpage.